

Which physically means that we are assuming that relatively few states from near the extrema are required to build the total wavefunction. This is not inconsistent with the previous approximation that the dispersion is isotropic and parabolic in our region of interest. So we have

$$\begin{aligned}
 (E - E_0)u(0, \mathbf{r}) \int \frac{d\mathbf{k}}{(2\pi)^3} C(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} &= u(0, \mathbf{r}) \int \frac{d\mathbf{k}}{(2\pi)^3} C(\mathbf{k}) \frac{\hbar^2}{2m_{eff}} \mathbf{k}^2 e^{i\mathbf{k} \cdot \mathbf{r}} + V_{slow} u(0, \mathbf{r}) \int \frac{d\mathbf{k}}{(2\pi)^3} C(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} \\
 (E - E_0) \underbrace{\int \frac{d\mathbf{k}}{(2\pi)^3} C(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}}}_{F(\mathbf{r})} &= -\frac{\hbar^2}{2m_{eff}} \nabla^2 \underbrace{\int \frac{d\mathbf{k}}{(2\pi)^3} C(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}}}_{F(\mathbf{r})} + V_{slow} \underbrace{\int \frac{d\mathbf{k}}{(2\pi)^3} C(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}}}_{F(\mathbf{r})} \\
 \therefore \left[-\frac{\hbar^2}{2m_{eff}} \nabla^2 + V_{slow} \right] F(\mathbf{r}) &= (E - E_0) F(\mathbf{r})
 \end{aligned} \tag{5.9}$$

For a impurity Coulomb potential, it leads to the *hydrogen atom like*-

$$\left[-\frac{\hbar^2}{2m_{eff}} \nabla^2 + \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{e^2}{r} \right] F(\mathbf{r}) = (E - E_0) F(\mathbf{r}) \tag{5.10}$$

This forms the basis of the hydrogenic impurity model. It also tells us when that works and what are its limitations. You can try to construct the equation if the band is not isotropic, as an exercise. The $F(r)$ function that we introduced is not really a wavefunction, in the sense that it is not the solution of the original Schrodinger equation. The wavefunction is Ψ , which we defined at the beginning and can be constructed if we know $F(r)$.

The justification of a similar model for indirect (and multiple valley) semiconductors like Si is more involved (this was given by Luttinger). Also the simple form of the equation will need modification if the band is not isotropic. The summary of the result is that for direct gap single valley (GaAs) the shallow donor level cluster around one number. See the figure 5.1. For indirect gap Si, the clustering does not happen so well. We will in general take this numbers as experimentally determined parameters.

5.2 A few useful numbers about Si, Ge and GaAs

5.3 Fermi Level in an intrinsic (undoped) semiconductor

If the material is undoped, then all the electrons in the conduction band (CB) must have been thermally excited from the valence band (VB). This fact is sufficient to tell us where the (E_f) should be. The electron and hole densities must be,

$$n = \int_{E_C}^{\infty} dE D(E) f(E) \tag{5.11}$$

$$p = \int_{-\infty}^{E_V} dE D(E) (1 - f(E)) \tag{5.12}$$

Let us assume that the dispersion relations are very simple

$$E_e(k) = E_C + \frac{\hbar^2 k^2}{2m_e} \tag{5.13}$$

$$E_h(k) = E_V - \frac{\hbar^2 k^2}{2m_h} \tag{5.14}$$

where E_C , E_V denote the bottom and the top of the conduction and valence bands respectively

Table 5.1: List of commonly used parameters of Silicon, Germanium and Gallium Arsenide

	Silicon	Germanium	Gallium Arsenide
Atoms cm^{-3}	5.0×10^{22}	4.4×10^{22}	4.4×10^{22}
Crystal structure	Diamond	Diamond	Zincblende
Density (gm cm^{-3})	2.33	5.33	5.32
Dielectric constant	11.9	16	13.1
Electron affinity (eV)	4.1	4.0	4.1
Effective density of states in conduction band at 300K: $N_c(\text{cm}^{-3})$	2.8×10^{19}	1.04×10^{19}	4.7×10^{17}
Effective density of states in valence band at 300K: $N_v(\text{cm}^{-3})$	1.04×10^{19}	6.0×10^{18}	7.0×10^{18}
Band gap at 300K (eV)	1.12	0.66	1.42
Intrinsic carrier concentration at 300K: $n(\text{cm}^{-3})$	1.5×10^{10}	2.4×10^{13}	1.8×10^6
electron effective mass : in units of m_0 , the free electron mass	0.98, 0.19	1.64, 0.082	0.067
hole effective mass : in units of m_0 , the free electron mass	0.16, 0.49	0.044, 0.28	0.082, 0.45
Intrinsic electron mobility at 300K ($\text{cm}^2\text{V}^{-1}\text{s}^{-1}$)	1350	3900	8500
Intrinsic hole mobility at 300K ($\text{cm}^2\text{V}^{-1}\text{s}^{-1}$)	480	1900	400
Electron diffusion coefficient at 300K (cm^2s^{-1})	35	100	220
Hole diffusion coefficient at 300K (cm^2s^{-1})	12.5	50	10

The density of states (in 3D), including spin degeneracy is then given by:

$$D(E) = \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} (E - E_C)^{1/2} \quad \text{for} \quad E > E_C \quad (5.15)$$

$$D(E) = \frac{1}{2\pi^2} \left(\frac{2m_h}{\hbar^2} \right)^{3/2} (E_V - E)^{1/2} \quad \text{for} \quad E < E_V \quad (5.16)$$

PROBLEM : Calculate the density of states in 2D and 1D for parabolic bands. Explain why it is allright to take one of the limits to be infinity in equations 5.11 & 5.12, even though all bands have finite extents.

Now to evaluate equations 5.11 & 5.12 we proceed as :

$$\begin{aligned} n &= \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} \int_{E_C}^{\infty} dE (E - E_C)^{1/2} \frac{1}{e^{\beta(E - E_F)} + 1} \\ &= \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} \frac{1}{\beta^{3/2}} \int_0^{\infty} du \frac{u^{1/2}}{e^u e^{-\beta(E_F - E_C)} + 1} \quad \text{where} \quad u = \beta(E - E_C) \\ &= 2 \left(\frac{2\pi m_e k_B T}{h^2} \right)^{3/2} \left(\frac{2}{\sqrt{\pi}} \int_0^{\infty} du \frac{u^{1/2}}{e^u e^{\beta(E_C - E_F)} + 1} \right) \end{aligned} \quad (5.17)$$

$$(5.18)$$

Now we identify the integral within the brackets as a Fermi-Dirac integral, defined as :

$$F_j(z) = \frac{1}{\Gamma(j+1)} \int_0^{\infty} dx \frac{x^j}{e^z e^x + 1} \quad (5.19)$$

Further the "effective density of states" in the conduction band is defined as

$$N_C = 2 \left(\frac{2\pi m_e k_B T}{h^2} \right)^{3/2} \quad (5.20)$$

Note however that the dimension of N_C is not the same as $D(E)$. With the definitions eqn 5.19 & 5.20, eqn 5.17 then reduces to

$$n = N_C F_{1/2} \left(\frac{E_C - E_F}{k_B T} \right) \quad (5.21)$$

You can prove that the number of holes is given by

$$p = N_V F_{1/2} \left(\frac{E_F - E_V}{k_B T} \right) \quad (5.22)$$

The Fermi-Dirac integrals appear often in physics. They are tabulated as "special functions". We can show that if E_F is reasonably *below* E_C , such that $\frac{E_F - E_C}{k_B T} < -4$ the integral is very closely approximated by $e^{\frac{E_F - E_C}{k_B T}}$. This is called the non-degenerate regime where the electron and hole densities are given by

$$n = N_C e^{\beta(E_F - E_C)} \quad (5.23)$$

$$p = N_V e^{\beta(E_V - E_F)} \quad (5.24)$$

For charge neutrality we must have $n_i = p_i$ for undoped (intrinsic) semiconductors only. Multiplying eqns 5.23 & 5.24

$$\begin{aligned} n_i^2 &= n_i p_i = N_C N_V e^{-\beta(E_C - E_V)} \\ \therefore n_i &= \sqrt{N_C N_V} e^{-\beta E_g/2} \end{aligned} \quad (5.25)$$

Clearly the intrinsic carrier density falls rapidly with increasing band gap. We can compare the result with the data in table 5.1.

Note however that the product of the carrier densities is independent of the location of the Fermi level even when $n \neq p$. This is a very important fact and allows us to write

$$np = n_i^2 \quad (5.26)$$

even when the source of the charges are dopants. In such cases (we will see in the next section) the Fermi level moves away from its intrinsic position. The electron and hole densities can then become vastly unequal - but they do so in such a way that the np product still remains the same. We now solve eqns 5.23 & 5.24 for E_f and get the intrinsic Fermi level (E_{fi})

$$E_{fi} = \frac{E_C + E_V}{2} + \frac{3}{4} k_B T \ln \frac{m_h}{m_e} \quad (5.27)$$

PROBLEM : Show that the deviation of the electron density (n) from intrinsic density (n_i) and the deviation of the Fermi level (E_f) from the intrinsic Fermi level (E_{fi}) are related as :

$$n = n_i e^{\beta(E_f - E_{fi})} \quad (5.28)$$

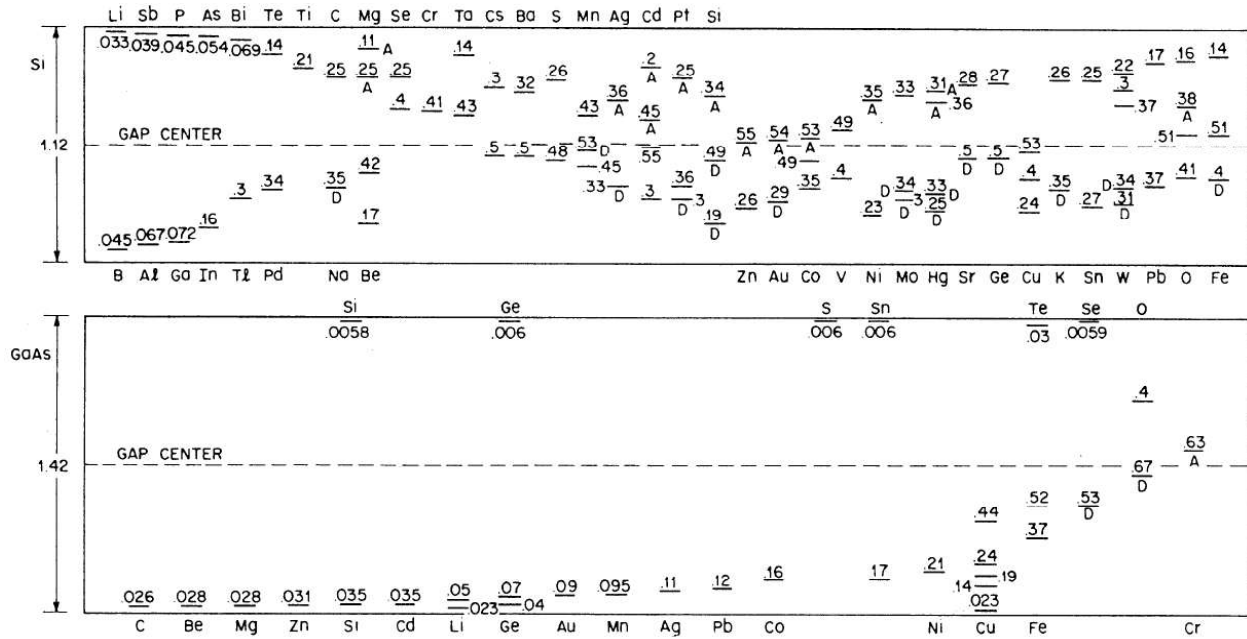


Figure 5.1: Impurity levels in Silicon and Gallium Arsenide (taken from the book by S.M. Sze). Notice that the shallowest levels tend to cluster somewhat around a value.

5.4 Fermi level in a doped semiconductor

We now come to the more practical situation, where there are dopants and ask: where is the Fermi level? If there are dopants then n and p are no longer equal. In fact the number of carriers supplied by ionised dopants can be several orders larger than the intrinsic carrier densities.

The fundamental point is that all the atoms of the host lattice and the dopants were initially neutral. But inside the semiconductor there are now four sources of charge :

- Negatively charged electrons in the conduction band (n)
- Unoccupied (positively charged) ionised donor atoms (N_d^+)
- Negatively charged ionised acceptor atoms (N_a^-)
- Unoccupied states (holes) in the valence band

The sum total of all these four must continue to be zero. To start with the valence band was full and the conduction band was empty (intrinsic semiconductor), then we put in neutral donor atoms (capable of giving out an electron) and neutral acceptor atoms (capable of capturing an electron). So the sum total must remain zero.

Thus if we can write down the carrier concentrations in the conduction and valence band and calculate the fraction of dopants which are ionised (as a function of E_f) then we can have an equation where E_f is the only unknown. This is how one determines the location of E_f

$$n + N_a^- = p + N_d^+ \quad (5.29)$$

We know how to write n and p as a function of E_f using equations 5.23 and 5.24. Now we ask, what is the probability that a donor will ionise. This question is an interesting exercise in statistical physics. The donor site (*e.g* Phosphorous in Silicon) can exist in four states

1. It may lose its electron (charge = +1 , energy = 0)

2. It may be occupied by a spin up electron (charge = 0, energy = E_D)
3. It may be occupied by a spin down electron (charge = 0, energy = E_D)
4. It may be occupied by one spin up and one spin down electron (charge = -1, energy = $2E_D + U$ where "U" is the large repulsive energy cost of putting two electrons on the same site, making the state very improbable.)

The dopant densities are not very large compared to the density of atoms of the host lattice. It is rarely more than 1 in 10^3 to 10^4 . So we can treat each dopant atom in isolation and the electron can be localised on the atom.¹ Each dopant can exchange electrons with the "sea" of conduction band electrons. It is thus in equilibrium with a larger system and can exchange particles with it - thus its temperature and chemical potential must be the same as that of the larger system.

So we write the partition function as (with μ , the chemical potential set as E_f)

$$\begin{aligned}
 Z_G &= \sum_{E,N} e^{\beta(E - \mu N)} \\
 &= e^{-\beta(0-0)} + e^{-\beta(E_D - E_f)} + e^{-\beta(E_D - E_f)} + e^{-\beta(2E_D + U - 2E_f)} \\
 &\approx 1 + 2e^{-\beta(E_D - E_f)}
 \end{aligned} \tag{5.30}$$

The mean occupancy (probability that the dopant is *not* ionised) is then,

$$\begin{aligned}
 1 - \frac{N_D^+}{N_D} &= 0.P(0) + 1.P(\uparrow) + 1.P(\downarrow) + 2.P(\uparrow\downarrow) \\
 &= \frac{2e^{-\beta(E_D - E_f)}}{Z_G} \\
 &= \frac{2e^{-\beta(E_D - E_f)}}{1 + 2e^{-\beta(E_D - E_f)}} \\
 &= \frac{1}{\frac{1}{2}e^{\beta(E_D - E_f)} + 1}
 \end{aligned} \tag{5.31}$$

Note that this is not simply a Fermi-Dirac distribution. The fraction of ionised donors is

$$\frac{N_D^+}{N_D} = \frac{1}{1 + 2e^{-\beta(E_D - E_f)}} \tag{5.32}$$

The similar expression for the fraction of ionised (negatively charged) acceptors is

$$\frac{N_A^-}{N_A} = \frac{1}{1 + 4e^{-\beta(E_f - E_A)}} \tag{5.33}$$

The factor 4 is a result of the fact that the electron sitting on the acceptor could have come from four possible places - spin up/down from heavy hole band, spin up/down from light hole band. The split off band does not come into the picture because it is too far down.

One type of dopant only

If we neglect the valence band and the acceptors (which can be justified if only donors are present), combining eqns 5.23 and 5.32 we get

¹However if the dopant densities *are* very high then the dopant states will not be localised. This condition is called "Mott transition". It is among the most studied problems in semiconductor physics.

$$N_C e^{\beta(E_f - E_C)} = \frac{N_D}{1 + 2e^{-\beta(E_D - E_f)}} \quad (5.34)$$

E_f is the only unknown in eqn 5.34 and can be solved (numerically if required).

PROBLEM : Show that the carrier density can now be obtained by solving the following equation: (which is in turn obtained by using eqn 5.34)

$$n^2 + nN_C \frac{e^{-\beta\Delta}}{2} - N_D N_C \frac{e^{-\beta\Delta}}{2} = 0 \quad (5.35)$$

where $\Delta = E_C - E_D$.

The fermi level can be obtained by solving

$$x^2 + x \frac{e^{-\beta\Delta}}{2} - \frac{N_D}{N_C} \frac{e^{-\beta\Delta}}{2} = 0 \quad (5.36)$$

where $x = e^{\beta(E_f - E_C)}$

If you put $N_D = 0$ in either of the two equations you would get an unphysical answer. Why is this so?

PROBLEM :

In a system with N_D donors, N_A acceptors, N_D^+ donors and N_A^- acceptors are ionised. Each donor (acceptor) level has a degeneracy of g_D (g_A). There are n electrons in the conduction band and p holes in the valence band. (In general $g_D = 2$, but g_A may be different from 2.). Then

$$N_D^+ = \frac{N_D}{(g_D n / N_C) \exp \beta(E_C - E_D) + 1} \quad (5.37)$$

And the corresponding result for the acceptors:

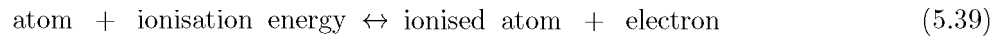
$$N_A^- = \frac{N_A}{(g_A p / N_V) \exp \beta(E_A - E_V) + 1} \quad (5.38)$$

Here N_C and N_V are the conduction and valence band effective density of states which have been defined earlier. Notice that the fermi energy does not appear in these relations.

A semiconductor may be doped with both (acceptors and donors) types of dopants. In a situation where there are a large number of donors and a few acceptors (*i.e* $N_D \gg N_A$), how would eqn 5.35 (the previous problem) be modified?

5.4.1 Thermal ionisation (Saha equation) of the dopant system

It is instructive to calculate the fraction of ionised dopants in another way. We can think of the problem as a thermal ionization of bound states - in a way that is very similar to the method of calculating the ratio of ionised to unionised atoms (of a certain species) in a hot plasma. We want to find the "chemical equilibrium point" of the reaction:



A certain fraction of atoms will exist in the dissociated state and a certain fraction will remain in the undissociated state. The fraction which minimises the free energy of the entire system (at a certain temperature) will be the equilibrium point.

Taking this approach we can calculate the ratio N_D^+/N_D by minimising the free energy of the entire system of free electrons and the dopants. First we write the free energy so that the free electron concentration n is the only variable.

$$\begin{aligned}
 F_{system} &= F_{electrons} + F_{dopants} \\
 F_{electrons} &= -kT \ln \frac{z^n}{n!} \text{ where for a single electron} \\
 z &= \sum_{\text{all states}} e^{-\beta E} \\
 &= V \frac{2}{h^3} \int d^3\mathbf{p} e^{-\frac{\beta p^2}{2m}} \\
 &= 2V \left(\frac{2\pi m kT}{h^2} \right)^{3/2}
 \end{aligned} \tag{5.40}$$

Now since $N_D - n$ dopant sites are occupied we have for the internal energy (U) and entropy (S)

$$U = -\Delta(N_D - n) \tag{5.41}$$

$$S = k \ln \left(2^{N_D - n} \frac{N_D!}{n!(N_D - n)!} \right) \tag{5.42}$$

$$F_{dopants} = U - TS \tag{5.43}$$

PROBLEM : Minimise $F_{system} = F_{electrons} + F_{dopants}$ w.r.t n , using Stirling's approximation for factorials as needed and show that you get exactly the same result as eqn 5.35. This is essentially a variant of the "Saha ionisation" equation, applied to a situation where the atoms and ions are not mobile, but only the electrons are.

5.4.2 General method of solving for the Fermi level

Consider a situation where a semiconductor is doped with N_D donors and N_A acceptors. We want the general solution for the location of E_F and all the carrier densities, ionisation probabilities. Since the semiconductor is overall neutral we have using the charge neutrality condition

$$n + N_A = p + N_D^+ \tag{5.44}$$

$$N_C F_{1/2} \left(\frac{E_C - E_F}{k_B T} \right) + \frac{N_A}{1 + g_A e^{\beta(E_A - E_F)}} = N_V F_{1/2} \left(\frac{E_F - E_V}{k_B T} \right) + \frac{N_D}{1 + g_D e^{\beta(E_F - E_D)}} \tag{5.45}$$

- here $g_A = 4$ and $g_D = 2$ are the acceptor and donor degeneracies. E_A and E_D are the acceptor and donor levels.
- Since E_F is the only unknown here, we can plot the LHS and RHS by treating E_F as an independent variable. The position where they intersect must be the solution. Figure ... illustrates the situation.
- Notice that E_F is temperature dependent.
- Once E_F is determined all the quantities can be determined. In general this cannot be done analytically. An example on numerical solution is given in Fig 5.2, 5.3.

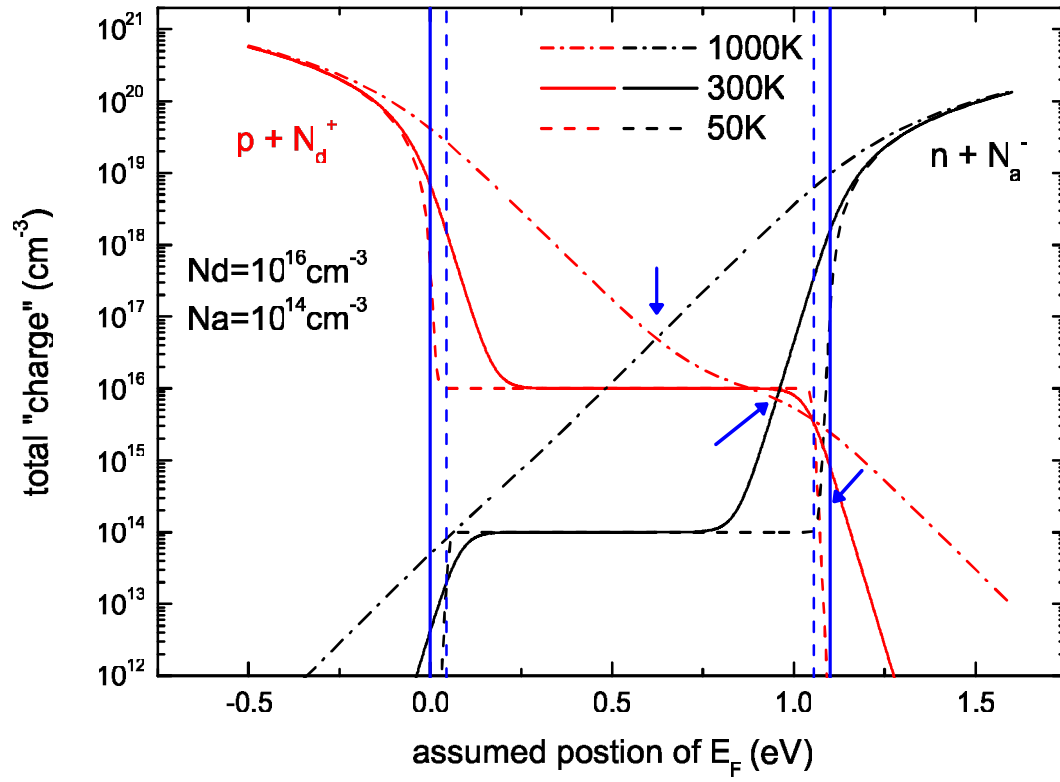


Figure 5.2: The LHS and RHS of eqn 5.45 are plotted for different temperatures. The intersection point is the solution for E_F .

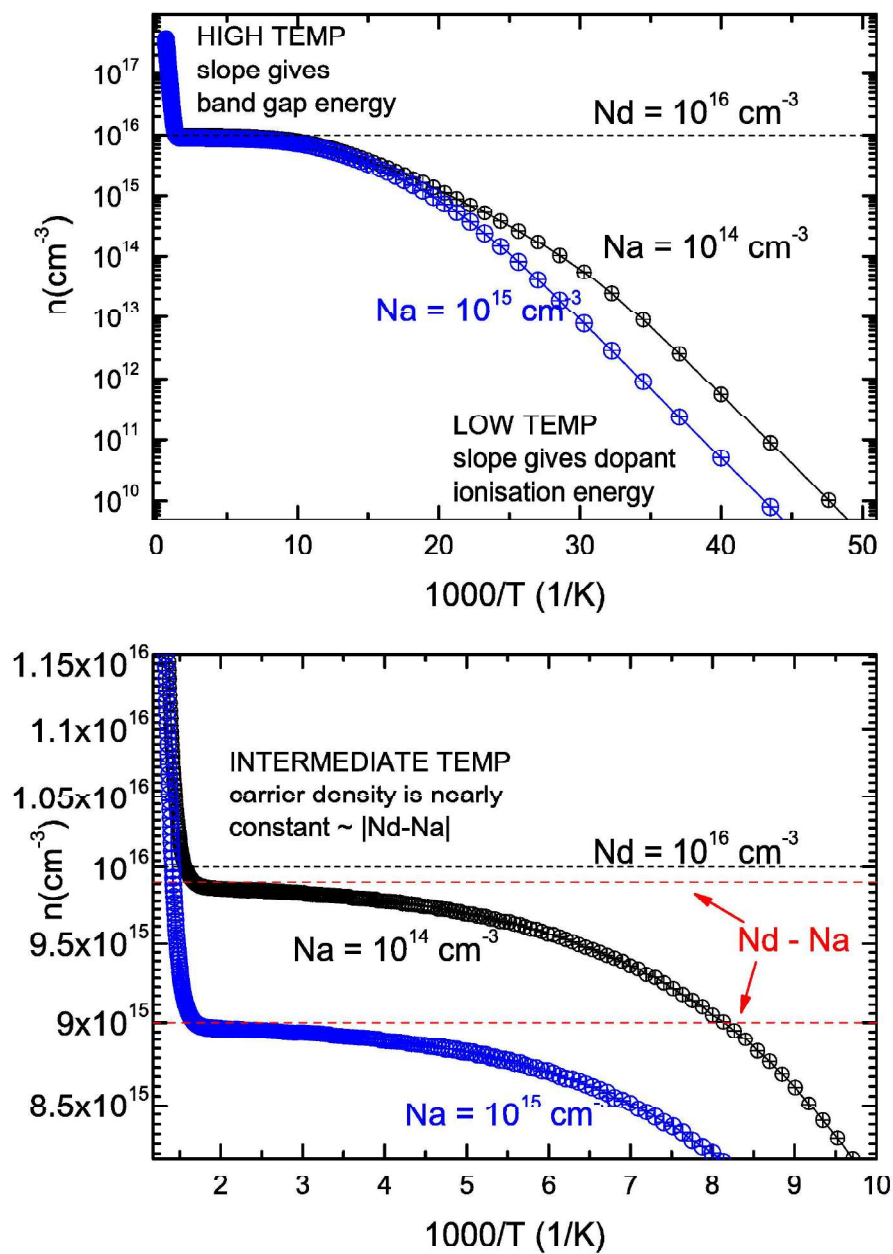


Figure 5.3: Figure shows the resulting carrier densities obtained from $n(T)$. Notice the various regimes.

5.5 The concept of a hole

We have seen earlier how an electron in a Bloch state behaves. In semiconductors we often encounter situations where a band is almost completely full but has a few vacant states. This happens because the Fermi level in an undoped semiconductor typically lies between the top of a filled band and the bottom of an unfilled band. We make the following observations and develop the method of dealing with this in steps. In what follows we denote the electron's wavevector by $\mathbf{k}_e$.

Remember that a hole is NOT a particle with a positive charge! It is one of the major conceptual mistakes.

1. The sum total of all the wavevectors in a band is

$$\sum_{\text{all states}} \mathbf{k} = 0$$

$$\therefore \sum_{\text{all states}} \mathbf{k} = -\mathbf{k}_e \quad (5.46)$$

$$(5.47)$$

if one state $\mathbf{k}_e$ is vacant. This "missing" wavevector is assigned to a "hole" and we say that the wavevector of the hole state is

$$\mathbf{k}_h = -\mathbf{k}_e \quad (5.48)$$

PROBLEM : Using this condition, explain why a hole state would tend to move in a $\mathbf{k}$ -space direction opposite to that of an electron when an electric field is applied? At no stage should you invoke anything to do with a positive charge.

- 2.

$$\sum_{\text{all states}} \mathbf{v}_g = 0 \quad (5.49)$$

Since $\mathbf{v}_g = \frac{1}{\hbar} \nabla E(\mathbf{k})$ and $E(\mathbf{k} + \mathbf{G}) = E(\mathbf{k})$ the result follows. We are essentially integrating the derivative of a periodic function over one period. This must be zero. This result is a *crucial* one as well.

3. Since one electron is missing from the band the energy of a hole should be

$$E_h = -E_e \quad (5.50)$$

4. Using equation 5.48 and eqn. 5.50 we arrive at the conclusion

$$\mathbf{v}_h = \mathbf{v}_e \quad (5.51)$$

The hole state moves with the same group velocity as the electron state.

5. The equation of motion (Lorentz force) for an electron state can be written down and *converted* to an equation for the hole as follows

$$\begin{aligned} \frac{d\mathbf{k}_e}{dt} &= -|e|(\mathbf{E} + \mathbf{v}_e \times \mathbf{B}) \\ \therefore -\frac{d\mathbf{k}_h}{dt} &= -|e|(\mathbf{E} + \mathbf{v}_h \times \mathbf{B}) \\ \therefore \frac{d\mathbf{k}_h}{dt} &= |e|(\mathbf{E} + \mathbf{v}_h \times \mathbf{B}) \end{aligned} \quad (5.52)$$

The hole state evolves as if it contains a particle of charge $|e|$. *This emerges from the equation of motion and was not put in by hand. There is no justification of assigning a positive charge to a*



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